

LA 3.2

Daniel Ransdorf

Jan Kodes

 V je v.pr. spojitých IR funkcí na $[1,5]$

(2) $\langle , \rangle$ definujeme $\langle f, g \rangle = \int_1^5 f(x)g(x)dx$

Určete ON bázi $W = \text{span}\{1, x, x^2\}$, $B = (\vec{u}_1, \vec{u}_2, \vec{u}_3)$

$$\vec{u}_1 = \frac{1}{\|1\|} = \frac{1}{\sqrt{4}} = \frac{1}{2} \quad \|\vec{1}\| = \int_1^5 1 dx = 5-1=4$$

$$\vec{u}_2 = \frac{\vec{v}_2 - \langle \vec{u}_1, \vec{v}_2 \rangle \vec{u}_1}{\|\vec{v}_2 - \langle \vec{u}_1, \vec{v}_2 \rangle \vec{u}_1\|} = \frac{x - \frac{1}{2} \int_1^5 x dx \cdot \frac{1}{2}}{\|x - \frac{1}{2} \int_1^5 x dx \cdot \frac{1}{2}\|} = \frac{x-3}{\|x-3\|} = \frac{\sqrt{3}(x-3)}{4}$$

$$\|x-3\| = \int_1^5 x^2 - 6x + 9 dx = \frac{16}{3}$$

$$\vec{u}_3 = \frac{\vec{w}_3}{\|\vec{w}_3\|} \text{ kde } \vec{w}_3 = \vec{v}_3 - \langle \vec{u}_1, \vec{v}_3 \rangle \vec{u}_1 - \langle \vec{u}_2, \vec{v}_3 \rangle \vec{u}_2$$

$$\vec{w}_3 = x^2 - \int_1^5 \frac{1}{2} x^2 dx \cdot \frac{1}{2} - \int_1^5 \frac{\sqrt{3}}{4} (x-3) x^2 dx \cdot \frac{\sqrt{3}}{4} (x-3) =$$

$$= x^2 - \frac{31}{3} - 8\sqrt{3} \cdot \frac{\sqrt{3}}{4} (x-3) = x^2 - \frac{31}{3} - 6x + 18 = x^2 - 6x + \frac{23}{3}$$

$$\vec{u}_3 = \frac{x^2 - 6x + \frac{23}{3}}{\|x^2 - 6x + \frac{23}{3}\|} = \frac{\sqrt{45}(x^2 - 6x + \frac{23}{3})}{\sqrt{256}}$$

Báze B tudíž je $\underline{\vec{u}_1 = \frac{1}{2}} \quad \underline{\vec{u}_2 = \frac{\sqrt{3}}{4}(x-3)} \quad \underline{\vec{u}_3 = \frac{\sqrt{45}}{\sqrt{256}}(x^2 - 6x + \frac{23}{3})}$

b) Orthogonalní projekce $\sin x$ na $\text{span}\{\vec{u}_1, \vec{u}_2, \vec{u}_3\}$

$$ax^2 + bx + c = \langle \vec{u}_1, \sin x \rangle \vec{u}_1 + \langle \vec{u}_2, \sin x \rangle \vec{u}_2 + \langle \vec{u}_3, \sin x \rangle \vec{u}_3$$

Podle Wolfram-Alpha dostáváme

$$\langle \vec{u}_1, \sin x \rangle \vec{u}_1 = \frac{1}{2} \int_1^5 \sin(x) \frac{1}{2} dx = 0,6416 = \frac{1}{4}(\cos 1 - \cos 5)$$

$$\langle \vec{u}_2, \sin x \rangle \vec{u}_2 = \frac{\sqrt{3}}{4} (x-3) \int_1^5 \frac{\sqrt{3}}{4} (x-3) \sin x dx = \frac{3}{16} (-2\cos 5 + \sin 5 - 2\cos 1 + \sin 1) (x-3)$$

$$\langle \vec{u}_3, \sin x \rangle \vec{u}_3 = \frac{45}{256} \left(\frac{4}{3} (\cos 1 - \cos 5) + 4 (\sin 5 - \sin 1) \right) \left(x^2 - 6x + \frac{23}{3} \right)$$

Výsledná aproximace je

$$ax^2 + bx + c = \underline{\underline{0,10x^2 - 0,94x + 1,69}}$$

c) norma rozdílu $\|\sin x - 0,1x^2 + 0,94x - 1,69\|$ se

$$\text{spočítá jako } \sqrt{\int_1^5 \left(\sin x - \frac{1}{10}x^2 + \frac{94}{100}x - \frac{169}{100} \right)^2 dx} = \underline{\underline{0,839}}$$